Highlights

Certified extreme-value analysis of significant wave height: maximum-likelihood fits and return levels over a public global hindcast, and environmental contours decided exactly[⋆]

Carlos Toledo

- Each extreme-value fit is proved the likelihood's one maximum in a recorded box.
- 46,602 fits over a public hindcast; 5,751 refused with the proof of a limit.
- A library's default call matches none of 60 certified fits; location fixed, 45.
- 150 benchmark contours read as exact polygons; 1.8 million states classified.
- 173 of 176 printed contour counts reproduced; 30 polygons cross themselves.

Certified extreme-value analysis of significant wave height: maximum-likelihood fits and return levels over a public global hindcast, and environmental contours decided exactly

Carlos Toledo[a,*]

[a] *Independent researcher, the cert-machine project,*

Abstract

A metocean design basis rests on two computations that are usually left to an optimiser or to a floating-point test: the marginal fit a return level is read from, and the environmental contour that sea states are counted against. This paper decides both. For the marginals it repeats the extreme-value method of Reis et al. (2026) — six families fitted by maximum likelihood to the unfiltered series and to daily, weekly and monthly maxima, ranked by four goodness-of-fit criteria, with 100- and 1000-year levels — on the method's own public hindcast. A fit is accepted only when the Krawczyk operator proves that a box around it contains exactly one zero of the score and the Hessian is proved negative definite over the box, in outward-rounded interval arithmetic over every datum; criteria and levels are enclosures over the box, and a ranking is DECIDED only where the enclosures separate. Over 2,589 open-sea cells, 40,778 of 46,602 fits are certified maxima and 5,751 generalized-gamma fits are refused with the proof that the likelihood is highest at the lognormal limit; a threshold in place of that proof changes the selected family at 544 of 7,767 cell-blocks. A scientific library's default

*Draft for review, version 0.2 of 2026-10-09. Version 0.1 (2026-10-01) held the marginal fits only; this version adds the environmental contours, condensed from the full re-decision of the contour benchmark (Toledo, 2026b), and leaves the general method — the arithmetic, the verdicts and the verification discipline — to Toledo (2026a). The author list and the CRediT roles are to be agreed with the group that published the reference method. Every number in this manuscript is read at build time from the records named in the Data availability statement (repository commit aae61990d737); the prose is authored and should be reviewed like any other.

*Corresponding author.

Email address: carlos@carlostoledo.co (Carlos Toledo)

call is the maximum it reports on none of 60 fits, and 19 of twenty rows of a published design table are off the maximum of the nearest public cell. For the contours it re-decides the environmental-contour benchmark whose three buoys are certified above: each of 150 submitted contours is read as the exact polygon its digits denote and 1,840,824 hourly sea states are classified inside, outside or on it with every sign decided. 173 of 176 printed counts are reproduced to the integer, two are traced to files rewritten after the count, 30 polygons cross themselves, and the expected numbers are certified enclosures. The novelty claimed is not interval arithmetic but the decision grammar — a verdict with a witness, refusal as a result — and a record any reader can re-run.

Keywords: significant wave height, extreme value analysis, return level, maximum likelihood, environmental contours, interval arithmetic, exact geometric computation, metocean design

---

1. Introduction

A metocean design basis states a significant wave height with a return period, and that number comes out of a chain: a record is blocked, a distribution is fitted to the blocks, one distribution is chosen by a goodness-of-fit statistic, and its tail is inverted at the return period (Coles, 2001; DNV, 2021; ISO, 2015). Each link is a computation, and each is usually carried out by a general-purpose optimiser whose stopping point is then called "the fit". Reis et al. (2026) applied this chain over a global wave hindcast: six parametric families — normal, lognormal, Weibull, exponentiated Weibull, generalized gamma and Gumbel — fitted by maximum likelihood to the unfiltered 3-hourly series and to daily, weekly and monthly block maxima, ranked by the Kolmogorov–Smirnov, Anderson–Darling, chi-square and mean-squared-error criteria, with the Anderson–Darling statistic chosen as the selector and 100- and 1000-year levels read off the selected family. Their conclusion, that the choice of criterion and of block size both move the selected distribution and the long-period return level, is a statement about the arithmetic of the method as much as about the sea. Beside the marginal stands the joint description of the sea states: an environmental contour, a closed curve in the plane of two metocean variables along which a structure is analysed for a given return period (Winterstein et al., 1993; Haselsteiner et al., 2017; Mackay and Haselsteiner, 2021). A contour is compared with data by counting the observations that fall outside it, and the benchmark that compared contour methods on common data (Haselsteiner et al., 2021)

— on three of the same buoys this paper certifies marginals for — made that count with a floating-point point-in-polygon test.

This paper asks a narrower question of both computations: at each step, is the number what the method says it is? A maximum-likelihood estimate is a zero of the score equations at which the Hessian is negative definite; a goodness-of-fit statistic is a sum over the data; a ranking is a comparison of such sums; a return level is a quantile of the chosen law; and the number of observations outside a contour is an integer. Each is decidable in finite arithmetic with rigorous error bounds, and the tools for doing so — interval arithmetic with outward rounding (Moore et al., 2009), the Krawczyk operator for the existence and uniqueness of a zero (Krawczyk, 1969; Rump, 2010), and validated global optimisation (Hansen and Walster, 2004; Tucker, 2011) — are decades old. So is exact geometric computation behind a floating-point filter, which decides every sign of a geometric predicate (Fortune and Van Wyk, 1996; Shewchuk, 1997; Yap, 1997). What has not been done, to our knowledge, is to apply them to the whole chain of an extreme-value study and to record the result so that a reader can re-run it. The present work does that for the method of Reis et al. (2026) on their own data and for the contour benchmark on its own files, and adds a grammar for the outcome: every fit, every ranking, every count and every printed number receives one of a small set of verdicts, each with the quantity that justifies it, and refusal is a result that is counted, not an error that is dropped. The grammar, the arithmetic and the verification discipline behind every verdict are those of the cert-machine method (Toledo, 2026a); this paper describes only what is particular to extreme-value analysis.

Three motivations come from practice. First, the likelihood of several families used for H_s has no interior maximum on many series: the generalized gamma's supremum is at its lognormal limit (Prentice, 1974; Stacy, 1962), and the exponentiated Weibull's runs to a Gumbel-type regime or to a Fréchet corner; an optimiser then prints the last iterate as the fit, and nothing in its output says so. Second, the same library called two ways returns two different 100-year levels for the same data, because a location parameter makes the likelihood unbounded (Hill, 1963; Harper et al., 2011); which of the two is a maximum is a question of fact, and this paper decides it. Third, design tables are printed from hindcasts that cannot be re-run outside the organisation that produced them (Bhaskaran et al., 2023); a certificate at the nearest public cell measures the distance between the two records and says which of the printed digits are a maximum-likelihood fit of anything. Fourth, a floating-point point-in-polygon test has no answer for a point on the boundary, answers wrongly near degeneracy without saying

so (Kettner et al., 2008), and presumes that the file it is given denotes a polygon; whether a benchmark's printed counts are the counts of the files on record is therefore a question of fact, and this paper decides it.

The contribution is therefore not a new distribution, a new contour construction or a new estimate of the 100-year wave. It is (i) a certificate per fit, (ii) a decision rule for rankings in which refusal is explicit, (iii) proofs, rather than thresholds, for the two families whose supremum is at a limit, (iv) a verdict vocabulary for numbers printed elsewhere, (v) an exact count per contour, with every submitted file read as the polygon it denotes and ON as a third answer, and (vi) the public, re-runnable record of all of this, with the method paper's own regional claims and the benchmark's printed tables decided against it. Section 2 describes the data; Sections 3 and 4 the certificate of a fit and the decision of a contour count; Section 5 reports the results; Section 6 discusses what a certificate does and does not say, and what a standard for marginal fits and contours could take from it.

2. Data

2.1. The hindcast

The reference method was applied by Reis et al. (2026) to the Ifremer WAVEWATCH III hindcast GLOBMULTI_ERA5_GLOBCUR_01 (Accensi, 2021; Alday et al., 2021): a global 0.5° grid at 3-hourly resolution, 1993 to 2024, forced by ERA5 winds and the global currents of the same system, distributed under CC BY-SA 4.0. The whole significant-wave-height field was read from the monthly files by byte range, each chunk hashed as read, and two cuts were kept. The first is a lattice of 3,268 nodes — a 4° global lattice, a 1° lattice of the Brazilian margin from 6°N to 36°S, and thirteen named nodes — at each of which the largest 3-hourly value of every UTC day was kept, 11,688 days from 1993-01-01 to 2024-12-31. Of these nodes, 679 are touched by the model's own sea-ice field at some step of the 32 years; under ice the model damps the waves to millimetres rather than leaving a gap, so those series are partly the ice's and are recorded but not certified. The remaining 2,589 open-sea cells (1,810 on the global lattice, 823 on the Brazilian margin) are the atlas of Section 5.1. The second cut is the full 3-hourly series, 93,504 values each, at 13 nodes chosen for the regions the method paper names: the Campos, Santos, Espírito Santo, Pelotas, Potiguar and Foz do Amazonas basins on the Brazilian margin; the Drake Passage, the central Antarctic Circumpolar Current, the high southern latitudes and the northern sub-Antarctic; the Arabian Sea, the western Pacific south of Japan

and the extratropical North Atlantic. A cell is the series at its node, not an area mean; nothing is interpolated.

2.2. Three buoys

To decide numbers other people printed for the same data, the three NDBC buoys of the environmental-contour benchmark of Haselsteiner et al. (2021) — its datasets A, B and C, hourly, the provided and the retained years together, 175,320 values each over 22, 22, 23 years — are certified at five block sizes (hourly, daily, weekly, monthly, annual). The benchmark's appendix prints the H_s marginal that each contributing team fitted to the provided ten years, and a scientific library's fits to the same series were recorded as printed (Section 5.6).

2.3. The contour benchmark

Exercise 1 of the benchmark (Haselsteiner et al., 2021) holds six hourly datasets: A, B and C are the sea states (H_s, T_z) of the three buoys above, D, E and F the wind–wave states (u_{10}, H_s) at three points of the coastDat hindcast. Participants received a "provided" part of each (ten years for A–C, twenty-five for D–F), the organizers kept a "retained" part of the same length, and nine groups submitted 1- and 20-year contours for A–C and 1- and 50-year contours for D–F as lists of vertices — 11 contour sets and 150 files in all, one group having submitted three constructions. Together the provided and retained parts hold 175,320 hourly states for each of A–C and 438,288 for each of D–F, 1,840,824 in all. The organizers counted, for every contour, the observations outside it and printed the counts in two tables beside the number expected for the contour's definition, 176 printed numbers. The contour files, the organizers' scripts, the datasets' pin and the preprint's tables (transcribed digit for digit) are held at the benchmark repository's commit `a1561fe7`, 167 files verified by SHA-256.

2.4. The block rule

A block maximum is the largest value in its calendar block, with the UTC day, the ISO 8601 week, the calendar month and the calendar year as the blocks; the rule lives in one module used by the ledger, the atlas and the browser so that it cannot drift between them. The T-year return level of a block of b hours is $F^{-1}(1 - b/(8766\,T))$, with $365.25 \times 24 = 8766$ hours in a year, as the method paper defines it; it is left undefined where fewer than one block falls in the return period.

3. The certificate of a fit

3.1. Families and criteria

The six families are those of Reis et al. (2026, §2.1), written in the same parameterisation: the normal and the lognormal in (μ, σ); the Weibull $F(x) = 1-\exp[-(x/\lambda)^k]$; the exponentiated Weibull $F(x) = (1-\exp[-(x/\lambda)^k])^\alpha$ (Mudholkar and Srivastava, 1993); the generalized gamma $F(x) = P(\alpha, (x/\lambda)^c)$ with P the regularized incomplete gamma function (Stacy, 1962); and the Gumbel $F(x) = \exp(-\exp[-(x-\mu)/\beta])$. None carries a location parameter except the Gumbel's own; this is the form the method paper prints and, as Section 5.5 shows, the form a library's default call does not use. The four criteria are the Anderson–Darling statistic $A^2 = -n - \frac{1}{n}\sum_i (2i-1)[\ln F(x_{(i)}) + \ln S(x_{(n+1-i)})]$ (Anderson and Darling, 1954), the Kolmogorov–Smirnov distance $D = \max_i \max(i/n - F(x_{(i)}),\, F(x_{(i)}) - (i-1)/n)$, the mean squared error $\frac{1}{n}\sum_i [F(x_i) - F_n(x_i)]^2$ against the empirical distribution function with ties, and a chi-square statistic on $\max(8, \lceil \log_2 n \rceil + 1)$ equal-width bins over $[\min x, \max x]$ with expected counts $E_j = n[F(e_{j+1}) - F(e_j)]$ kept where $E_j \geq 5$ and undefined below two kept bins, as Reis et al. (2026, §2.2) describe. Lower is better for all four. A seventh family, the generalized extreme value law (Jenkinson, 1955), of which the Gumbel is the $\xi = 0$ member, is certified beside the six and never ranked among them (Section 3.5).

3.2. A fit as a box

A maximum-likelihood estimate is a zero of the score $\nabla_\theta \ell(\theta)$. The certifier finds a candidate in floating point — Nelder–Mead on the log-likelihood followed by a Levenberg–Marquardt ascent — and then proves two things about a box X around it, both in outward-rounded interval arithmetic in which every datum is enclosed by its two neighbouring doubles and every elementary operation rounds outward (Moore et al., 2009):

1. the Krawczyk operator $K(X) = \hat{\theta} - Y\,\nabla\ell(\hat{\theta}) + (I - Y\,H(X))(X - \hat{\theta})$, with Y an approximate inverse of the Hessian at the centre and $H(X)$ the interval Hessian over X, maps X strictly into its interior, so that X contains exactly one zero of the score (Krawczyk, 1969; Rump, 2010);
2. the Hessian is negative definite over X: the leading principal minors of $-H(X)$ are enclosed by Sylvester's criterion and proved positive over the whole box, or, where the box's Hessian is too ill-conditioned for its minors, positive at the candidate with every Hessian over the box proved nonsingular, which gives the same conclusion on a connected box.

The zero is then the likelihood's one maximum in X, and X is the certificate: its half-width is recorded (at most 1.3×10^{-6} in the parameters on the 3-hourly ledger, where the widest 100-year level enclosure is 5×10^{-5} m). Everything downstream is an interval extension over X: each criterion, each return level, and the log-likelihood itself, so that two families' values on the same data can be compared. The arithmetic is that of Toledo (2026a, §3); what is particular to likelihoods is the length of the sums, since a running interval sum rounds outward at every one of n additions and at $n = 175{,}320$ its width is a thousand times what the terms carry, so each endpoint stream is summed by the Sum2 algorithm with the error bound of Ogita et al. (2005) added outward. The gamma-function family ($\ln\Gamma$, ψ, ψ', P, P^{-1}) is enclosed by Binet's formula with the first omitted term as the bound, the series of P with a geometric tail, and bisection with certain sign for P^{-1}; the score and Hessian come from second-order forward differentiation over the same arithmetic, so that no derivative is written by hand.

What this does not prove is that no better maximum exists elsewhere in the family: that is the search's claim, as it is any optimiser's, and the paper says so wherever a fit is stated. It also does not say that the family is the law of the sea, nor how far another record of the same length would move the number; the sampling width is a statistical statement and is kept apart (Section 3.6).

3.3. Rankings as decisions

With every statistic an enclosure, a ranking of families under a criterion is DECIDED when the lowest enclosure lies wholly below every other, and REFUSED otherwise, naming the overlap. A family without a certified fit refuses the ranking — its maximum might exist and win — with one exception: the generalized gamma whose likelihood is proved highest at its lognormal limit (Section 3.4) is left out, named, and the lognormal, which is what its likelihood reaches, is ranked in its stead. A climb stopped at a family's declared boundary proves nothing and blocks the ranking; so does a statistic the arithmetic cannot enclose, and a chi-square bin whose expected count straddles five. DECIDED ranks the criterion's arithmetic on this sample; it is not a test between models. The rule is one function, called by the ledger and by the browser, and the atlas's regional claims are decided through it.

3.4. Families whose supremum is at a limit

As $\alpha \to \infty$ with $c \to 0$ the generalized gamma tends to the lognormal (Prentice, 1974), so its likelihood's supremum is at least the lognormal's

maximum, and on many series the likelihood rises all the way there. In Stacy's coordinates the ridge is too flat for any contraction argument. In Prentice's (μ, σ, Q), with $Q = 1/\sqrt{\alpha}$ and the likelihood written as a power series in Q with proved tails, $Q = 0$ is an ordinary point, and a maximum beside the limit contracts (1,996 such fits are certified in these coordinates on the atlas, 25 on the 3-hourly ledger). Where no maximum is found the family is refused at its edge only with a proof: over a box $B \times (0, q_1]$, with B a neighbourhood of the lognormal fit of the form $\hat{\mu} \pm k\hat{\sigma}/\sqrt{n}$, $\hat{\sigma}(1 \pm k/\sqrt{2n})$ for the widest k among $3, 1, 0.3, 0.1$ that proves it, the Q-component of the Prentice score is proved negative, so every member there is less likely than the lognormal at the same (μ, σ) and hence than the lognormal's maximum. Like a certified maximum this is a local fact, and its neighbourhood is recorded. On buoy A's hourly series, for instance, the proof holds with $k = 3$ and $q_1 = 0.05$, the score's Q-component at most -5326 over the box.

The exponentiated Weibull's α runs away on another kind of series. With $\theta = \lambda^k$ and $\beta = \theta \ln \alpha$ the family reads $F(x) = \exp\{e^{\beta/\theta} \ln(1 - e^{-x^k/\theta})\}$, a Gumbel law of H_s^k with location β and scale θ once $e^{-x^k/\theta}$ is small over the data; the runaway α is a regime that the (α, k, λ) coordinates place in the millions or beyond, not an edge, and in (k, θ, β) the maximum is an ordinary point, certified there at 219 atlas cell-blocks. Where the climb runs on even in those coordinates, toward $k \to 0$ — a Gumbel law of $\ln H_s$, which is a Fréchet law — the family is refused with that reason (53 atlas cell-blocks), and the ranking with it, because the family's supremum there is not yet proved to be the Fréchet limit itself. An earlier version of the ledger used a threshold ($\alpha > 500$ for the generalized gamma) in place of the boundary proof; on six 3-hourly series the family's maximum sits beside the limit at α between 951 and 2,323, certified there, and the threshold rule handed 17 of their 24 choices to another family. The present ledger keeps that rule's outcome beside the certificate's at every cell-block, so that the difference between a threshold and a proof can be counted (Section 5.1).

3.5. The seventh family

The method paper's six hold one extreme-value law, the Gumbel. The generalized extreme value law $F(x) = \exp\{-[1+\xi(x-\mu)/\sigma]^{-1/\xi}\}$ is certified beside them at every cell and block, written so that $\xi = 0$ is an ordinary point and with the support condition $1+\xi(x-\mu)/\sigma > 0$ enforced on every datum. Its likelihood has no global maximum — below $\xi = -1$ it is unbounded as the support's upper end reaches the largest datum — so its fit is the certified maximum in the region where the estimator is regular, $\xi > -0.5$ (Smith, 1985), and a box that reaches -0.5 is refused. The GEV is never

ranked among the six: the tables that read against the method paper are the paper's; a separate column records what the Anderson–Darling statistic decides among the seven by the same rule.

3.6. Statistical, not certified

A certificate says what a fit *is* on these 32 years; it does not say how far another 32 years of the same sea could move it. The ledger carries that width beside the records, labelled: the delta method's 95% intervals on the decided family's 100- and 1000-year levels, taken on the logarithm of the level so that they stay positive, from the observed information at the fitted point in the certificate's own coordinates and the level's gradient by central differences — asymptotic, with successive blocks taken as independent and the family as right (Coles, 2001); the runs estimator of the extremal index of the daily maxima (Smith and Weissman, 1994); and, on request in the browser, a profile-likelihood interval. These are statements of another kind and are drawn apart from the certified ones in every figure of this paper.

3.7. Printed fits

A number someone printed is decided over the box its digits allow: a printed "0.0634" is the interval $[0.06335, 0.06345]$, and the full log-likelihood and the levels are evaluated over that box against the certified fit on the same values. The verdict is REPRODUCED when the certified box lies inside the printed digits (they are its rounding); CONSISTENT when the digits lie inside the certified box; OFF THE MAXIMUM when the two are apart and the printed likelihood lies below the certified maximum by a proved margin; NOT THE CERTIFIED FIT when they are apart without such a margin; OUTSIDE ITS SUPPORT when the printed location leaves data with zero density; NOT A MEMBER when a printed parameter lies outside the family; and NOT DECIDED otherwise. The words are the same for the benchmark's marginals, the library's fits and the design table.

3.8. The record and its controls

The discipline is the method's (Toledo, 2026a, §5); its instances here are these. Every certifying module is pinned by its SHA-256 in the atlas ledger (10 modules), every worker of the atlas run refuses other bytes, and the page that serves the atlas refuses to build over a ledger its own bundle did not write; the corpus is pinned the same way (digest ef375fbfe509). A battery of 1,678 checks runs before any page is written, among them red controls that must fire: logarithms skewed to the right refuse the generalized gamma at its limit with the proof; a 40,000-point generalized gamma with $\alpha \approx 900$ is

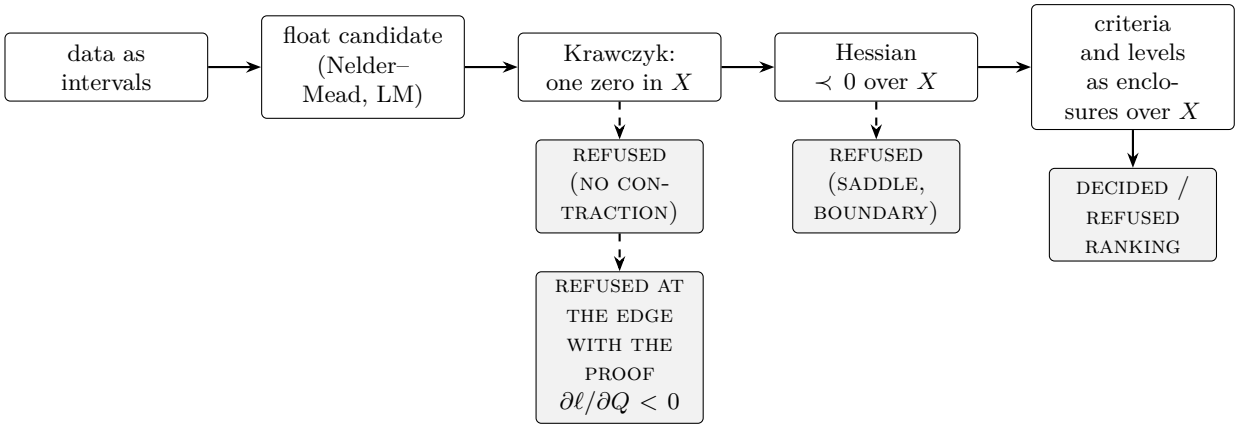


Figure 1: The certificate of one fit. A float search only proposes; the Krawczyk operator and the second-order test decide; every statistic is an interval extension over the certified box X. A refusal is a recorded result with its reason, and the generalized gamma can be refused at its lognormal limit only with the boundary proof of Section 3.4.

certified beside its limit, where a stop at $\alpha = 500$ would miss it; a family stopped at its boundary blocks the ranking; a saddle point is refused by the second-order test; and the printed-fit decision has 15 checks with 6 red controls of its own. Any cell can be re-certified in a browser from the pinned data by the same bytes, and the result is compared with the ledger's record by digest. The atlas took 3.9 hours on seven worker threads of a laptop; the 3-hourly ledger 1,186 seconds.

4. The decision of a contour count

4.1. The count the benchmark made

For each contour set and dataset, the organizers' script concatenates the provided and retained hours, reads the contour file in a column order hand-coded per participant, and asks matplotlib's `Path.contains_points` (Hunter, 2007) about every hour: a crossing-number (even-odd) test in double precision on the polygon closed implicitly from its last vertex to its first, with no defined answer on the boundary. An observation is outside when the test says it is not inside; a second count keeps those with $H_s > 1$ m (A–C), or $u_{10} > 1$ m/s and $H_s > 1$ m (D–F). The tables print, for the long return period, the mean over three datasets and the three integers, and for the one-year contours the three-dataset means to one decimal; beside each, the number expected for the contour's definition. That count is the benchmark's metric, and it is a finite, decidable quantity.

4.2. Literals and the polygon a file denotes

Every number in the benchmark is a decimal literal, and each is read as the rational its digits denote, beside its double image. A file's vertices define a polygon closed from the last vertex to the first, as the organizers' test closes it; for each polygon the record holds, exactly, whether any two non-adjacent edges cross properly, touch or fold back, which edges cross, whether the implicit closing edge is among them and how long it is, the signed area, and the maximum of H_s along the contour, which is attained at a vertex because H_s is linear on an edge. A polygon with no crossing, touch or fold is *simple*; the others are reported and counted all the same, because they were counted.

4.3. Inside, outside, on

A point is classified by the even-odd crossing number of a horizontal ray (Shimrat, 1962; Hormann and Agathos, 2001), in the half-open convention of the organizers' library. Every predicate is a sign — of the 2×2 determinant that orients a point against an edge, or of a difference of two literals — and each is evaluated first in double precision and trusted only when its magnitude clears a forward-error bound proved for the inputs (Higham, 2002; Fortune and Van Wyk, 1996): with every coordinate of magnitude at most M and $u = 2^{-53}$, the determinant's computed value is within $48uM^2$ of the true one, which is 8.7×10^{-11} at the $M = 128$ that covers the benchmark, so a computed determinant above 10^{-10} in magnitude has the right sign and one below is re-decided in arbitrary-precision integers, each literal scaled to sixty-four decimal places. A point on an edge or at a vertex is ON, a third answer the float test does not have; the nonzero-winding count is kept beside the even-odd one. A printed integer *agrees* with the record when it lies in $[n_{\text{out}}, n_{\text{out}} + n_{\text{on}}]$ and is *exact* when it equals n_{out}; a printed mean is read the same way through its rounding; a printed number that agrees under neither reading is *not the count of the file on record*.

4.4. The expected numbers

Beside each count the benchmark prints $E = n\,\alpha_t$, with $\alpha = 1/(T \times 8766)$ the hourly exceedance probability of a T-year contour and α_t the probability that the contour is exceeded anywhere: α itself for a total-exceedance construction, an exact rational, and for IFORM $\exp(-\beta^2/2)$ with $\beta = \Phi^{-1}(1-\alpha)$ (Mackay and Haselsteiner, 2021). The latter is enclosed: erfc by Laplace's continued fraction (Abramowitz and Stegun, 1964, §7.1.14), whose consecutive convergents bracket the value, β by bisection that tightens only

where the sign is certain, and the rest in interval arithmetic. The printed value is then read as a rounding of the enclosure, or not.

4.5. Controls

The decider is one module with no dependencies, so the page on which a reader classifies any sea state against any submitted contour runs the bytes that wrote the ledger. Its battery holds 218 checks, 14 of them red controls that must fire: among them a determinant, found by random search, whose double-precision value is $+3.6 \times 10^{-15}$ and whose exact sign is negative; a bow-tie that must be reported as one crossing; a square traced twice, on which the even-odd and winding rules must part; the two columns swapped, which must move a planted point across the contour; and a point 10^{-20} outside an edge, which must not be ON. The full re-decision is in Toledo (2026b).

5. Results

5.1. The atlas

Table 1 and Fig. 2 give the selection by the Anderson–Darling statistic over the 2,589 open-sea cells at the three block sizes the atlas holds. Of the 46,602 fits, 40,778 (87.5%) are certified maxima; 5,751 (12.3%) are generalized-gamma fits refused at the lognormal limit with the proof; 53 are exponentiated-Weibull climbs that run toward the Fréchet corner in both coordinate systems; and 20 are refused for other reasons. The choice is DECIDED at 2,569 cells on the daily maxima, 2,573 on the weekly and 2,541 on the monthly, and REFUSED at 20, 16 and 48. The exponentiated Weibull is the decided choice at 1,597 cells on the daily maxima and 1,115 on the monthly; the Weibull at 9 and 69. The threshold rule of Section 3.4 — the generalized gamma past $\alpha = 500$ and the exponentiated Weibull past 10^4 taken for "the limit" and left out — names a different family, or none, at 544 cell-blocks (7.0%): 167 daily, 137 weekly, 240 monthly (Fig. 3).

The selection the statistic makes is not a sanity check. The decided family's 100-year level lies below the largest daily maximum of the 32-year record at 2,607 of the 7,663 decided cell-blocks with a level (34%); the median ratio of the level to the record is 1.06, the ninetieth percentile 1.29, and it exceeds two at 61 cell-blocks and five at 11 (Fig. 4). The largest is at 15°N, 62°E in the Arabian Sea on the daily maxima, where the exp. Weibull is the decided choice and its 100-year level is 128.5 m against a record of 6.87 m, 18.7 times the record. Table 2 gives the node's full 3-hourly series: the exponentiated Weibull is decided at every block in its Gumbel coordinates,

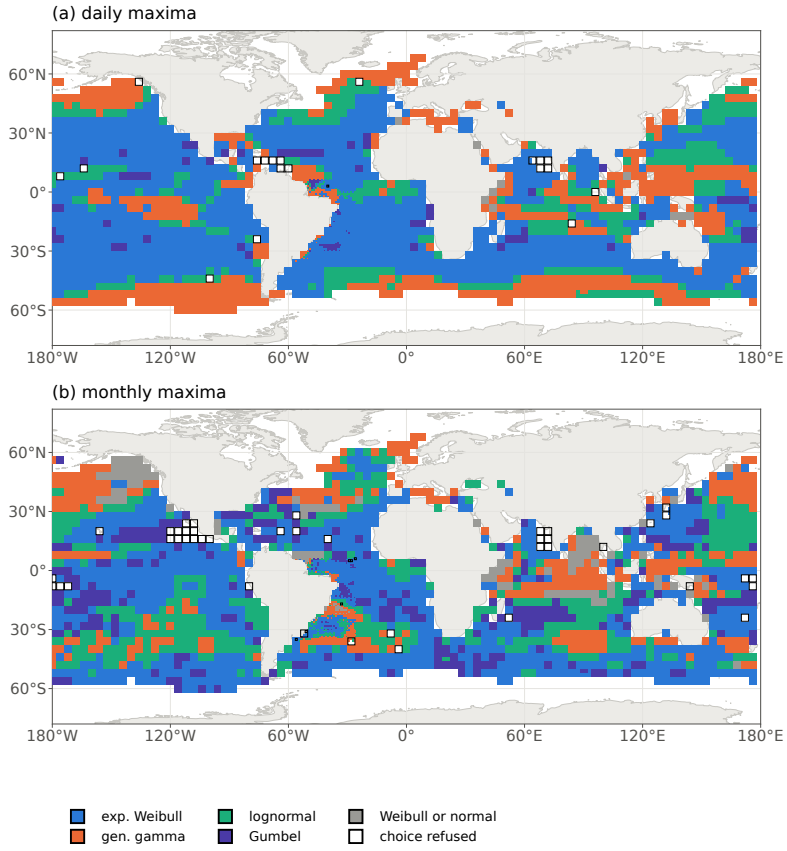


Figure 2: The family the Anderson–Darling statistic selects among the six at every open-sea cell, (a) daily and (b) monthly maxima; a hollow square is a refused choice. Cells are drawn as squares centred on their node: 4° on the global lattice, 1° on the Brazilian margin. The Weibull and the normal, selected at few cells, share one colour.

Table 1: The family the Anderson–Darling statistic selects among the six, by block, over the 2,589 open-sea cells: with the certificate (left) and with the threshold rule of Section 3.4 in place of the boundary proof (right). A refused cell is one where the enclosures overlap or a family has no certified fit.

certificate	daily	weekly	monthly
normal	2	9	52
lognormal	355	437	520
Weibull	9	13	69
exp. Weibull	1,597	1,452	1,115
gen. gamma	495	473	455
Gumbel	111	189	330
refused	20	16	48

threshold rule	daily	weekly	monthly
normal	2	9	52
lognormal	477	520	610
Weibull	9	13	69
exp. Weibull	1,619	1,435	1,041
gen. gamma	353	388	361
Gumbel	116	220	445
refused	13	4	11

with a 100-year level of 360.9 m on the unfiltered series and 23.6 m on the monthly maxima against a record of 6.87 m. The certificate proves what the method yields, including where the method yields a number no engineer would use: the two-mode monsoon climate of that sea puts the family's maximum in a regime where the fitted law is a Gumbel law of a small power of H_s, and nothing in the selection rule objects. A design standard built on this method would have to say so.

5.2. The method paper's regional claims

Section 3 of Reis et al. (2026) describes its maps in words: where a family gains or loses as the block grows, and where the return levels fall or rise with it. Ten of those sentences speak of the blocks the atlas holds, and each was read as a box (a latitude–longitude region or the global lattice), the blocks it names and a rule over the certified cells there (Appendix Appendix A gives the quotations). A cell counts for a family only where the choice is DECIDED; a refused cell counts for nobody and could count for anybody, so a claim HOLDS when it is true however the refused cells fell, DOES NOT HOLD when it is false however they fell, and is UNDECIDED when they could make it either. Table 3 gives the result: 6 hold, 2 do not, 2 are undecided. The two that do not hold concern the generalized gamma in the Southern Ocean.

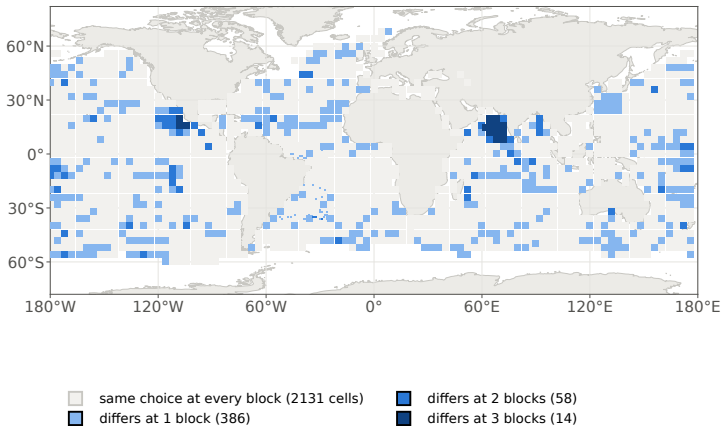


Figure 3: Where a threshold rule in place of the boundary proof names a different family, or none, at one, two or all three of the daily, weekly and monthly blocks.

Table 2: The Arabian Sea node (15°N, 62°E): the family the Anderson–Darling statistic selects, its 100-year level, the exponentiated Weibull's shape k in its Gumbel coordinates, and the largest value on record, by block. On the 3-hourly ledger the exponentiated Weibull is certified at every block in the coordinates (k, θ, β) of Section 3.4.

block	selected	100-year H_s (m)	k	record (m)
3-hourly	exp. Weibull	360.9	0.032	6.87
daily	exp. Weibull	128.5	0.050	6.87
weekly	exp. Weibull	46.3	0.101	6.87
monthly	exp. Weibull	23.6	0.226	6.87

The two undecided ones are the global rise of the Weibull with block size and the same transition in the monsoon Indian Ocean, where the refused cells are enough to turn either way. The same ten claims were decided a second time with the threshold rule of Section 3.4 in place of the boundary proof (the table's sixth column): 6 hold, 3 do not and 1 are undecided under that rule, and 4 of the ten (claims 3, 5, 6, 10) change verdict between the two readings — which is to say that the difference between a threshold and a proof reaches the sentences a reader takes from the maps. A reader who disagrees with a box can redraw it: the atlas decides any region typed or drawn in the browser by the same function.

Table 3: The ten regional claims of Reis et al. (2026, §3) decided over the certified cells ("verdict"), and decided again with the threshold rule of Section 3.4 in place of the boundary proof ("threshold"). A count "decided" is the number of cells where the named family (or the named relation between the two blocks' levels) is DECIDED; "refused" the cells where the choice is refused; "of" the cells in the box.

#	region	cited	blocks	verdict	threshold	counts
1	the global 4° lattice	§3.1, Fig. 2	daily, weekly, monthly	UNDECIDED	UNDECIDED	exp. Weibull 977 → 888 → 728 (refused 19, 14, 42, of 1810); Weibull 9 → 11 → 61 (refused 19, 14, 42, of 1810)
2	south of South America (40–58° S, 80–56° W)	§3.2	daily	DOES NOT HOLD	DOES NOT HOLD	exp. Weibull 12, gen. gamma 9, lognormal 1 of 22 (refused 0)
3	the Southern Ocean (40–65° S)	§3.2	daily, weekly	DOES NOT HOLD	HOLDS	gen. gamma 166 → 131 (refused 1, 3, of 382)
4	the Southern Ocean (40–65° S)	§3.2	monthly	HOLDS	HOLDS	exp. Weibull 227, lognormal 73, Gumbel 48, gen. gamma 32, normal 1 of 382 (refused 1)
5	the monsoon Indian Ocean (0–25° N, 45–80° E)	§3.2 (Indian Ocean)	daily, monthly	UNDECIDED	HOLDS	Weibull 1 → 3 (refused 5, 6, of 44)
6	the western Pacific south of Japan (24–36° N, 124–148° E)	§3.2 (western Pacific)	weekly, monthly	HOLDS	DOES NOT HOLD	exp. Weibull 16, Gumbel 4, gen. gamma 2, Weibull 1 of 26 (refused 3)
7	the North Atlantic (40–64° N, 60° W–0°)	§3.3	daily, monthly	HOLDS	HOLDS	lower at 38, not at 21, refused 3, of 62
8	the North Pacific (40–60° N, 140° E–130° W)	§3.3	daily, monthly	HOLDS	HOLDS	lower at 70, not at 9, refused 2, of 81
9	the Arabian Sea (5–25° N, 50–75° E)	§3.3	daily, monthly	HOLDS	HOLDS	lower at 16, not at 0, refused 7, of 23
10	the western Pacific south of Japan (24–36° N, 124–148° E)	§3.3	daily, monthly	HOLDS	DOES NOT HOLD	higher at 17, not at 6, refused 3, of 26

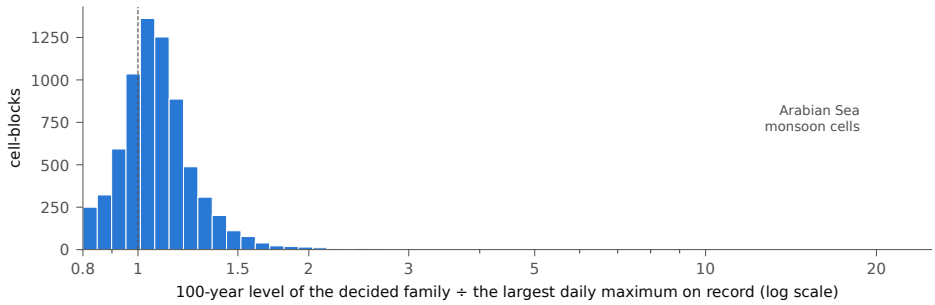


Figure 4: The decided family's 100-year level divided by the largest daily maximum on record, over every decided cell-block of the atlas (daily, weekly and monthly maxima). The dashed line is the record itself.

5.3. The seventh family

The GEV is certified at 7,765 of 7,767 atlas cell-blocks and refused at 2 where its box reaches $\xi = -0.5$. Its shape is certified below zero — a finite upper end, which none of the six families can have — at 4,925 cell-blocks, and above zero at 2,840 (Fig. 5a); no certified box straddles zero. Among the seven, the Anderson–Darling statistic names the GEV at 1,489 cell-blocks (456 daily, 541 weekly, 492 monthly; Fig. 5b). On the 1° Brazilian margin it is named at 697 cell-blocks, at 592 of them with ξ certified below zero. At the 45 cell-blocks where the exponentiated Weibull runs toward its Fréchet corner the GEV is certified at 45, its ξ positive at 45 and its A^2 below every certified family's at 45 — the tail the corner points to — and the choice among seven stays refused at 45 of them, because the exponentiated Weibull's supremum there is not yet proved to be that limit. The GEV runs away where the six do: at the Arabian Sea node its daily 100-year level is of the same order as theirs (Table 2).

On the statistical side, kept apart: the decided family's 100-year level carries a delta-method 95% interval of 7% of its value at the median cell on the daily maxima, 12% on the weekly and 17% on the monthly; the runs estimator of the extremal index of the daily maxima runs from 0.16 to 0.68 over 2,589 cells, median 0.44, so that a design-life computation which takes successive daily maxima as independent counts more independent chances than the sea gives.

5.4. The 3-hourly series: buoys and nodes

On the three buoys at five blocks and the thirteen nodes at the method paper's four blocks, 353 of 402 fits are certified maxima, 47 are generalized-

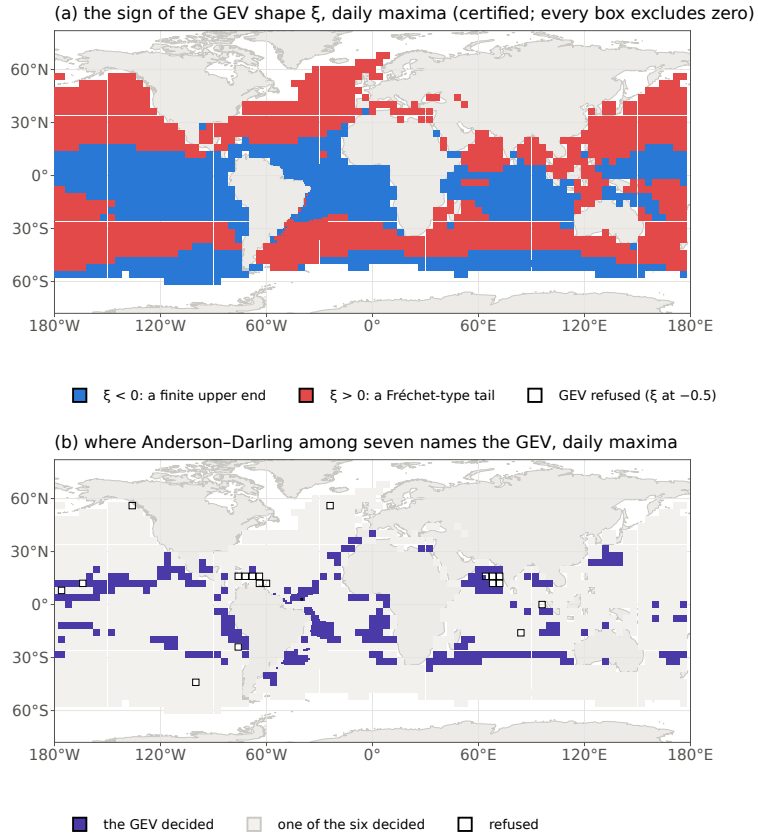


Figure 5: The seventh family on the daily maxima. (a) The sign of the GEV shape ξ, certified: below zero (a finite upper end) or above zero (a Fréchet-type tail); no certified box contains zero. (b) The cells where the Anderson–Darling statistic, applied to the seven families by the rule of Section 3.3, names the GEV.

Table 4: The 3-hourly series on which the generalized gamma's certified maximum lies beside its lognormal limit ($\alpha > 500$ in Stacy's coordinates, certified in Prentice's), and how many of the four criteria's choices move when the family is left out as a threshold rule would.

node	block	α	choices moved
campos	monthly	1,287	3
foz-amazonas	weekly	951	2
drake	daily	1,670	4
acc-central	native	1,370	4
acc-central	daily	2,323	1
north-atlantic	weekly	2,315	3

gamma fits refused at the limit with the proof, and 2 — the exponentiated Weibull on buoys A's and C's annual maxima — are refused where the climb runs toward $k \to 0$. The Anderson–Darling choice is DECIDED on 65 of 67 series; the two refused are those annual series. Read against the method paper's three findings: "the exponentiated Weibull performs best for high-frequency data" is decided for it on buoy A's hourly series and against it on B's (lognormal) and C's (gen. gamma), and on the hindcast's unfiltered 3-hourly series it is the decided choice at 6 of 13 nodes, the lognormal winning Campos and Santos; "the traditional Weibull becomes more appropriate as the block size increases" is decided against on every series — the Weibull is the decided best of 0 of the 12 buoy series and 0 of the 52 node series at the paper's blocks; and "the choice of goodness-of-fit test significantly influences the selected distribution" holds, the four criteria naming different families on 2 of 12 buoy series and 20 of 52 node series. Reading only the family the statistic selects at each of the four blocks, the 100-year level at buoy B runs from 8.10 m (weekly) to 12.21 m (native), a factor of 1.51; on the nodes the factor runs from 1.020 (southern-high) to 1.62 (south-of-japan) apart from the Arabian Sea (Appendix Appendix B). The generalized gamma's maximum sits beside its lognormal limit on 6 node series, at α between 951 and 2,323 (Table 4); certified there, it wins 17 of their 24 choices, every one of which a stop at $\alpha = 500$ hands to another family.

5.5. The same software, two calls

A scientific library's maximum-likelihood fitter (SciPy 1.18.1; Virtanen et al., 2020) was called on every buoy block series two ways and its output recorded as printed: with the location parameter fixed at zero, which is the families as the method paper writes them, and with the default call, in which the location is free. Each printed fit was then decided against

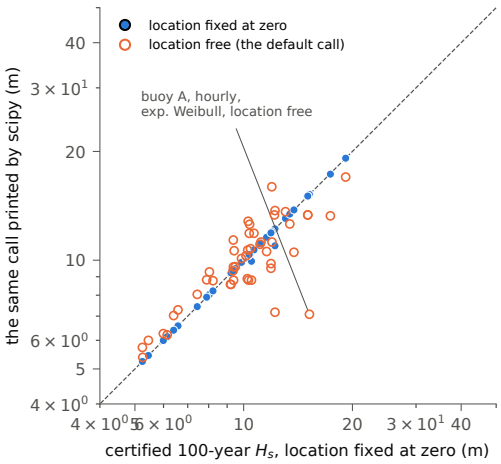


Figure 6: The 100-year level printed by the library against the certified maximum on the same block maxima, for the four families with a scale parameter on the three buoys at five blocks: location fixed at zero (filled) and the default call with a free location (hollow). The dashed line is equality.

the certificate by the rule of Section 3.7. With the location fixed, 45 of 60 fits agree with the certificate to half a centimetre in the 100-year level; 10 are generalized-gamma fits printed below the limit the family's likelihood is proved to rise to; 3 are off the maximum; 2 are the exponentiated Weibull where the certificate refuses it. With the location free, no fit agrees: 52 cannot be decided at all, because a location parameter below the smallest datum makes the likelihood unbounded (Hill, 1963) and the printed point is a stationary point of a function with no maximum; 7 lie below a member of their own family at location zero, by a proved margin; and 1 places a Weibull location above one of buoy A's monthly maxima, which then has zero density (Table 5). The pair that matters for a design basis is buoy A's hourly exponentiated Weibull: the fixed-location call returns a 100-year level of 15.20 m and is the certified maximum; the default call returns 7.08 m from a point 19,850 log-likelihood units below that member of its own family (Fig. 6). Across the default calls the printed 100-year level differs from the certified one by between -8.12 and +4.01 m. This is not a defect of the library, which does what it is asked; it is a property of the likelihood, documented for the three-parameter Weibull by Harper et al. (2011), and a reason for a certificate to travel with the number.

Table 5: The library's fits that are neither the certified maximum nor undecidable, with the printed 100-year level (its difference from the certified one in parentheses) and the proved log-likelihood deficit below the certified maximum or, for a location-free fit, below the family's member at location zero.

series	family	call	verdict	100-year H_s (m)	ℓ below
A hourly	exp. Weibull	location free	below a member	7.08 (−8.12)	1.98×10^4
A hourly	gen. gamma	location fixed	below its limit	8.35	2.53×10^3
A hourly	gen. gamma	location free	below a member	8.15	2.52×10^3
A daily	exp. Weibull	location free	below a member	13.34 (−1.70)	3.96
A daily	gen. gamma	location fixed	below its limit	8.15	1.67×10^2
A daily	gen. gamma	location free	below a member	6.47	7.76×10^2
A weekly	exp. Weibull	location fixed	off the maximum	17.31 (−0.07)	0.00
A weekly	gen. gamma	location fixed	below its limit	9.99	18.86
A monthly	Weibull	location free	outside support	11.37	–
A monthly	gen. gamma	location fixed	below its limit	11.42	1.25
A annual	gen. gamma	location fixed	below its limit	10.33	0.69
B hourly	gen. gamma	location fixed	below its limit	8.65	1.03×10^3
B daily	gen. gamma	location fixed	below its limit	7.35	64.35
B weekly	gen. gamma	location fixed	below its limit	8.03	5.86
B weekly	gen. gamma	location free	below a member	8.79	0.29
B monthly	gen. gamma	location fixed	below its limit	8.94	2.08
B monthly	gen. gamma	location free	below a member	9.74	0.36
C hourly	gen. gamma	location fixed	off the maximum	10.96 (−1.25)	60.79
C hourly	gen. gamma	location free	below a member	7.18 (−5.03)	1.41×10^3
C daily	gen. gamma	location fixed	off the maximum	9.93 (−0.58)	1.34
C annual	gen. gamma	location fixed	below its limit	10.19	0.73

5.6. Printed fits: a benchmark's marginals and a design table

The environmental-contour benchmark (Haselsteiner et al., 2021) gave its contributing teams the same ten years of the three buoys and printed, in its appendix, the H_s marginal each fitted; 15 printed rows can be read as a family and decided on the hours they were fitted to (Appendix Appendix C). For buoy A the printed marginals put the hourly 100-year level anywhere between 5.25 m and 14.53 m, while the certified maximum-likelihood fits of the Weibull, the lognormal and the exponentiated Weibull on the same 82,805 hours give 5.25, 12.11 and 15.90 m. One contribution's two-parameter Weibull for A and B is the certified maximum-likelihood fit to its last printed digit, and its row for C is exactly the certified Weibull fit of the zero-up-crossing period, not of H_s; another's translated Weibull leaves 10,426, 10,904, 9,410 observed hours of A, B and C below its printed location, where its density is zero; a third's three-parameter lognormal is a certified local maximum of a likelihood that has no global one. None of this

is visible in the printed digits. It is visible in the certificate.

A design table was decided the same way. Bhaskaran et al. (2023) print annual-maximum fits of H_s — the Gumbel by least squares, maximum likelihood and moments, and the GEV by maximum likelihood — for forty offshore wind lease areas, from a commercial hindcast that cannot be rerun outside it. For the five South Atlantic areas, the two nearest open-sea cells of the public hindcast (31 to 61 km from the printed coordinates) were blocked by the calendar year, the Gumbel and the GEV were certified on the 32 annual maxima, and the printed digits were decided against them. Of the 20 rows, 19 are OFF THE MAXIMUM and 1 is OUTSIDE ITS SUPPORT; the printed 100-year wave lies between −2.03 and +3.14 m from the certified fit's, outside the certified fit's 95% delta-method interval at 19 of 20 (Table 6, Fig. 7). This is the distance between two records — a commercial hindcast at the lease area and a public one at the nearest cell — and not an error in the table: at the Projeto Açu area the two nearest public cells, 31 and 77 km away, give Gumbel 100-year levels of 3.21 and 5.47 m. What the certificate adds is exactly that the distance is now a number, per row, with the method's own arithmetic on the public side decided rather than estimated.

Table 6: The South Atlantic rows of the design table of Bhaskaran et al. (2023) decided against the nearest open-sea cell of the public hindcast: the printed parameters (the GEV's shape with the sign convention of this paper), the 100-year level the printed digits imply, the certified fit's 100-year level and its 95% delta-method interval (statistical, not certified), their difference, and the verdict of Section 3.7.

area	method	printed	printed 100-yr	certified 100-yr	95% interval	difference	verdict
UY01	Gumbel, least squares	6.80, 0.71	10.09	8.38	[7.60, 9.25]	+1.71	off the maximum
UY01	Gumbel, maximum likelihood	6.83, 0.57	9.48	8.38	[7.60, 9.25]	+1.10	off the maximum
UY01	Gumbel, moments	6.82, 0.62	9.70	8.38	[7.60, 9.25]	+1.32	off the maximum
UY01	GEV, maximum likelihood	6.81, 0.56, -0.08	9.01	9.61	[6.83, 13.51]	−0.59	off the maximum
Projeto Açu	Gumbel, least squares	3.68, 0.40	5.55	3.21	[3.00, 3.44]	+2.34	off the maximum
Projeto Açu	Gumbel, maximum likelihood	3.68, 0.38	5.46	3.21	[3.00, 3.44]	+2.24	off the maximum
Projeto Açu	Gumbel, moments	3.69, 0.35	5.33	3.21	[3.00, 3.44]	+2.12	off the maximum
Projeto Açu	GEV, maximum likelihood	3.71, 0.40, 0.14	6.36	3.22	[2.76, 3.77]	+3.14	off the maximum
Farol wind	Gumbel, least squares	5.10, 0.59	7.84	5.86	[5.34, 6.42]	+1.99	off the maximum
Farol wind	Gumbel, maximum likelihood	5.12, 0.51	7.49	5.86	[5.34, 6.42]	+1.64	off the maximum
Farol wind	Gumbel, moments	5.12, 0.52	7.54	5.86	[5.34, 6.42]	+1.68	off the maximum
Farol wind	GEV, maximum likelihood	5.11, 0.51, -0.02	7.40	5.76	[4.94, 6.72]	+1.65	off the maximum
Sopros do RJ	Gumbel, least squares	2.84, 0.26	4.06	5.47	[5.01, 5.97]	−1.41	off the maximum
Sopros do RJ	Gumbel, maximum likelihood	2.85, 0.24	3.98	5.47	[5.01, 5.97]	−1.49	off the maximum
Sopros do RJ	Gumbel, moments	2.85, 0.23	3.94	5.47	[5.01, 5.97]	−1.54	off the maximum
Sopros do RJ	GEV, maximum likelihood	2.85, 0.24, 0.05	4.14	5.04	[4.53, 5.61]	−0.90	off the maximum
Projeto Ubu	Gumbel, least squares	2.16, 0.22	3.20	4.93	[4.49, 5.41]	−1.73	off the maximum
Projeto Ubu	Gumbel, maximum likelihood	2.17, 0.18	3.03	4.93	[4.49, 5.41]	−1.90	off the maximum
Projeto Ubu	Gumbel, moments	2.17, 0.19	3.07	4.93	[4.49, 5.41]	−1.85	off the maximum
Projeto Ubu	GEV, maximum likelihood	2.16, 0.17, -0.09	2.83	4.86	[4.07, 5.81]	−2.03	outside its support

5.7. The contour benchmark, decided

The printed counts. Of the 176 numbers printed in the benchmark's two tables, 173 are the exact count to the integer, or the exact mean to the digit printed (Fig. 8). The organizers' arithmetic is reproduced wherever the file

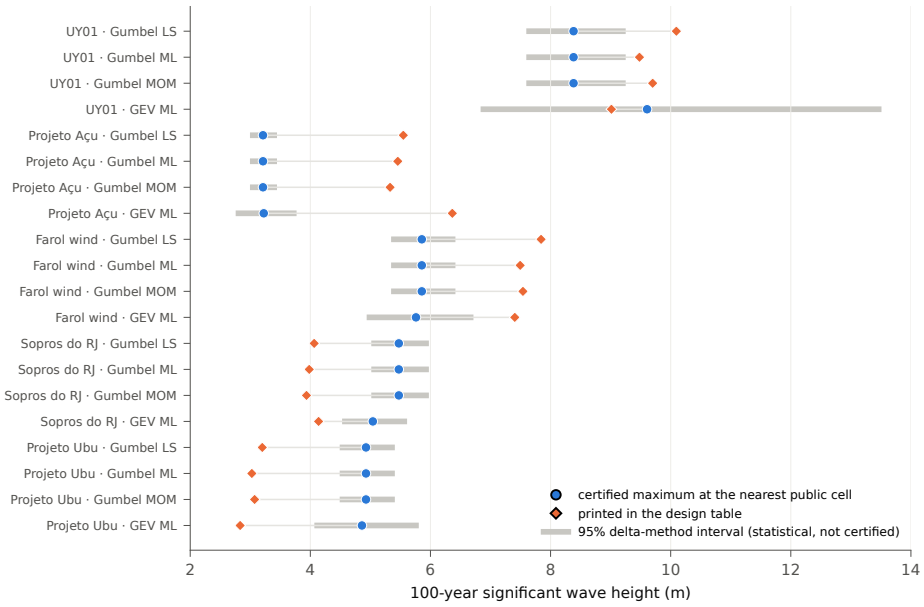


Figure 7: Twenty printed 100-year waves of a design table (Bhaskaran et al., 2023) against the certified fit at the nearest public cell, with the latter's 95% delta-method interval. The difference measures the distance between two hindcasts at two points; the certificate makes it a number per row.

they counted is the file on record; what the exact re-decision adds in those cells is the standing of the number as a decision, with the polygon's digest, the rule and the boundary points recorded beside it. One more number agrees only within the boundary points: two hourly observations lie exactly on contribution 2's one-year wind–wave contours, on E at 1989-06-17 20:00 and on F at 1986-06-27 20:00, where vertices printed with 3 decimals and observations printed with 4 coincide; the exact mean outside is 1185.7, the mean with both counted outside is 1186.3 — the printed value, and what the float test does — and the record says ON and reads the printed mean as consistent with either side. Two numbers, the one-year sea-state row of contribution 3, agree under neither reading — a mean of 104.0 outside against a printed 82.3 — and the benchmark repository's own history says why: the three files were added on 2020-08-11 and rewritten on 2020-09-25, removing 53, 41 and 55 rows in which a coordinate was `nan`. The printed row corresponds to no polygon that can now be built from the repository, and the decision refuses to reproduce it approximately; the verdict is a statement of provenance, not of error in the organizers' arithmetic.

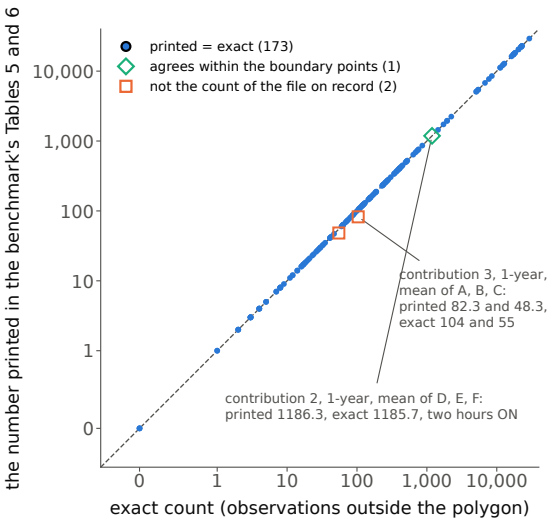


Figure 8: The 176 numbers printed in the contour benchmark's two tables against the exact counts (both axes symmetric-logarithmic). 173 lie on the diagonal; the one that agrees within the two boundary points and the two that are not the count of the file on record are marked.

The polygons. A point-in-polygon test presumes a polygon. 30 of the 150 files describe a closed polyline whose edges cross each other properly, up to 26 times, and 49 do not repeat their first vertex, so the region counted includes an edge, from the last listed vertex back to the first, that nobody listed (Fig. 9). Under the even-odd rule a self-crossing polygon still has an inside, but not the region a reader sees when the curve is drawn; both rules were applied to every observation on every contour, and no observation changes side between them, so the printed counts do not depend on the choice. What depends on it is what the number is a count of.

The expected numbers. The benchmark's expected numbers are certified: on the sea-state datasets the total-exceedance expectations 20 and 1 are exact, because the full datasets hold 20×8766 hours, and the IFORM values 197 and 11.5 (and 492 and 12 on the wind–wave datasets) are the roundings of certified enclosures.

Beyond the tables. Because every maximum along a contour is a literal read from the file, the highest contour can be set against the highest observation exactly. In dataset A the highest observed state, 11.7976 m at 2010-02-26

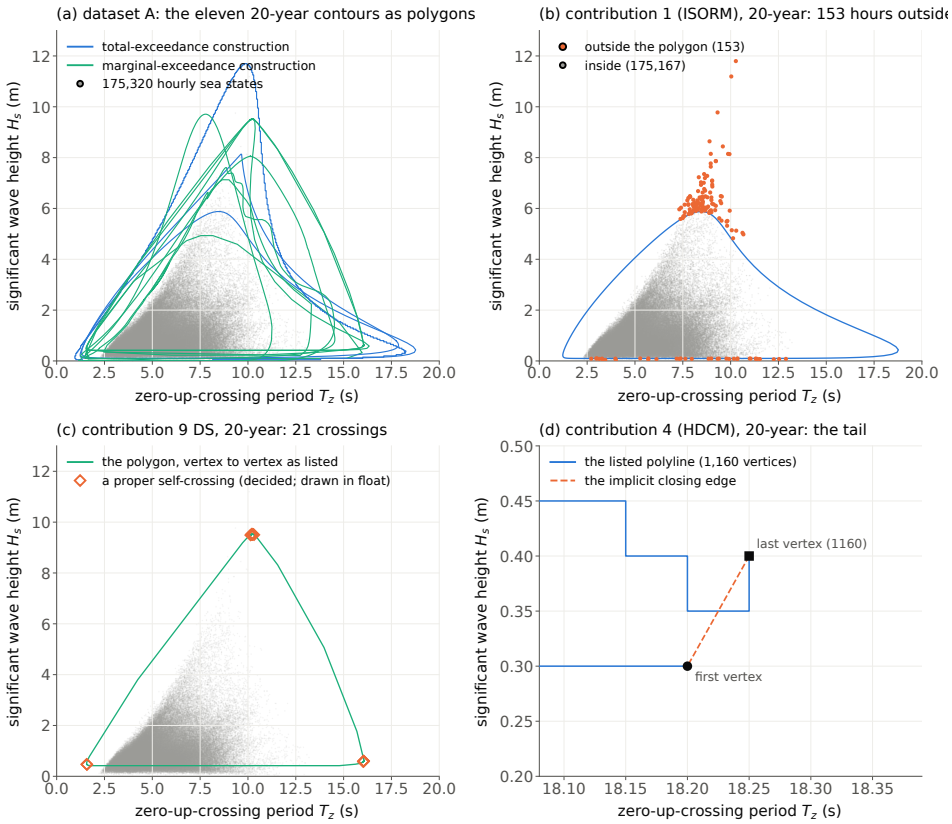


Figure 9: Dataset A of the contour benchmark. (a) The eleven 20-year contours as the polygons their files denote, each closed from its last vertex to its first, over the 175,320 hourly sea states. (b) Contribution 1's ISORM contour with the 153 hours decided outside it marked; the drawing refuses to build if matplotlib's own test on the same polygon does not reproduce the count. (c) Contribution 9's direct-sampling contour drawn vertex to vertex in file order, with its 21 proper self-crossings (the count is decided; the positions are drawn in floating point). (d) The tail of contribution 4's highest-density contour: the listed polyline ends at the square, and the implicit closing edge back to the circle, the longest edge of the polygon, crosses the staircase.

05:00, lies in the retained years and above the maximum of every one of the eleven 20-year contours, the highest of which, contribution 4 (HDCM), reaches 11.70 m and has exactly 1 observation outside it on the full dataset: that hour. Over the six datasets, 27 of the 150 contours reach above the highest observed wave. On the retained years alone — the years the contributors did not see — the highest-density contours are exceeded 12 times in all against an expectation of 3.08; these are counts, not tests, since hourly states are serially correlated (Haselsteiner et al., 2021).

The two halves of this paper meet on buoy A. The same benchmark's printed marginals put the hourly 100-year H_s anywhere between 5.25 and 14.53 m (Section 5.6), and its printed contour counts are exact in every cell whose file is on record: the counting was right wherever it could be checked, while the marginal digits beneath the joint models ranged over a factor of nearly three in the 100-year level and included one law whose density is zero at observed hours.

6. Discussion

What a certificate says. For each fit the record holds the digest of the data, the choices made (the family, the block rule, the coordinates), a box proved to contain the likelihood's one maximum, the enclosure of each statistic and level over it, and the digests of the code; for each ranking it holds the verdict and, where refused, the overlap; for each printed number it holds the box the digits allow and the verdict; and for each contour it holds the file's digest, the polygon the file denotes with its diagnostics, the rule and its boundary convention, the counts under both rules, and the hours of any boundary point. Anyone can check these claims from the record alone, in any arithmetic that rounds outward, and the atlas does so in a browser from the pinned data. In the regulatory sense nothing here certifies anything: offshore units are approved by classification societies under national rules, and a design basis is theirs to accept (DNV, 2021; ISO, 2015). The word is used in this paper in its mathematical sense only, for a statement with a machine-checkable proof, and the objects offered are such proofs.

What it does not say. A certificate does not rank models: DECIDED means that one criterion's arithmetic separates the families on this sample, and the Arabian Sea node shows a decided choice that no one would design to. It does not say that a better maximum is absent elsewhere in a family, which remains the search's claim; a branch-and-bound proof over the family would close that gap and is the natural next step. It does not give the sampling

width of a level, which is a statistical statement made beside the record and labelled; nor the hindcast's own error against the sea it simulates. And a decided regional claim depends on the box drawn for it; the atlas lets the method paper's authors redraw their own. On the contour side, no contour is called right or wrong: a count of exceedances is the benchmark's own metric, taken on its own terms; a self-crossing polygon is a property of a file as submitted, not necessarily of the construction that produced it; the enclosures certify the benchmark's formula for the expected number, not the independence it assumes; and the printed tables read are those of the preprint held in the benchmark's repository.

What it changes. Two practical consequences follow. The first is for software: the same library returns two 100-year levels for the same data depending on a default, neither of which announces whether it is a maximum, and the generalized gamma's likelihood has no interior maximum on 5,751 of 46,602 atlas fits, where any optimiser prints a point on the way. A pipeline that fixes the location and uses the boundary proof agrees with the certificate; one that does not can be a factor of two off at the 100-year level with no warning. The second is for standards. A standard for marginal fits — which the operators of the Brazilian margin's fields have reason to want, since every contractor builds its own model (Reis et al., 2026) — can be read off the atlas: which choices it must fix (the location, the block rule, the coordinates in which a family is fitted, the treatment of a family at its limit, the criterion), and where a certificate rather than an optimiser has to say what the fit is. The certificate is the second half of such a standard: the data stay where they are, and the proof travels. The same holds for contours, at no cost to whoever draws them: a contour submitted or delivered as an explicit polygon — closed, with no vertex repeated consecutively, no coordinate that is not a number, a header naming the columns — with the rule for "inside", the convention for a point on the boundary and the thresholds declared, and the count published as an integer per contour and dataset beside the file's digest, is decidable by construction; an exact point-in-polygon decider is a few hundred lines in any language and re-decides the whole benchmark in seconds. The expected number beside each count should be printed exactly where it is a rational and as an enclosure where it is not.

Cost. The atlas's 46,602 fits took 3.9 hours on a laptop's seven worker threads, with the generalized gamma's three climbs and the Prentice fallback the largest part; six fits of a 93,504-value 3-hourly series take minutes each.

Re-certifying one cell in a browser takes seconds. The contour benchmark is re-decided in 5.9 s on a laptop: 6.37×10^8 predicates, of which 237,173 (372 per million) fell below the filter's bound and were decided in integers. The arithmetic is JavaScript on IEEE-754 doubles with simulated directed rounding (one ulp outward per operation), which is conservative by a factor of about two in the last bit and costs nothing at the accuracy needed here.

7. Conclusions

The extreme-value chain of Reis et al. (2026) was repeated on its own public hindcast with every fit certified as the likelihood's one maximum in a recorded box, every statistic and level an enclosure, every ranking a decision or a refusal with its reason, and the two families whose supremum lies at a limit treated by proof rather than by threshold. Over 2,589 open-sea cells, 40,778 of 46,602 fits are certified and 5,751 are refused at the generalized gamma's lognormal limit with the proof; a threshold in place of that proof changes the selected family at 544 cell-blocks and the verdict of 4 of the method paper's ten regional claims. The same decision applied to other people's numbers finds that a library's default call is the maximum it reports on none of 60 fits, that the printed marginals of a benchmark span a factor of nearly three in the hourly 100-year level on one buoy, and that 19 of 20 rows of a published design table are off the maximum of the nearest public cell by a measured amount. On the contour side, the 150 contours of the benchmark on the same buoys were read as the exact polygons their files denote and 1,840,824 hourly sea states classified inside, outside or on each with every sign decided: the organizers' procedure reproduces 173 of their 176 printed numbers to the integer, one more within two observations that lie exactly on a contour, and two are traced to files rewritten after the count; 30 polygons cross themselves and 49 are closed by an edge nobody listed; the expected numbers are certified. What is proposed to the metocean community is not a different estimate of the 100-year wave or a different contour but a different object to accompany them: a record that any reader can re-run, in which refusal is a verdict.

Declaration of competing interest

The author declares no competing financial or personal interests. The method paper is cited as the reference method; its authors are being consulted on this manuscript. The contour benchmark's organizers and contributors have not been consulted on this draft.

Data availability

The records, the certifier and the figure and table sources are public in the cert-machine repository (`https://github.com/carlostoledo1891/cert-machine`, commit aae61990d737): the atlas ledger `certs/hseva-atlas.json` (generated 2026-10-22), the 3-hourly ledger `certs/hseva-ledger.json` (2026-10-21), the library's fits as printed `certs/hseva-scipy.json` (2026-09-26), the design-table audit `certs/design-table-audit.json` (2026-10-22), the extracted hindcast series under `corpus/ww3-grid` and `corpus/ww3-points` (CC BY-SA 4.0, as the hindcast), and the certifier under `instruments/hseva` and `instruments/interval` (MIT); for the contours, the ledger `certs/ecbench-ledger.json` (generated 2026-09-12), the pinned corpus `corpus/ec-benchmark` (the 150 contour files verbatim, the organizers' scripts, the preprint's tables transcribed, and the pin of the datasets, which are fetched from commit `a1561fe7` of the benchmark's repository and verified by digest) and the decider `instruments/ecbench` (MIT). The contour report is at `https://carlostoledo.co/reports/ec-benchmark.html`, and any sea state can be classified against any submitted contour, by the same bytes, at `https://carlostoledo.co/instruments/contours`. The atlas with every cell re-certifiable in the browser is at `https://carlostoledo.co/instruments/return-level-atlas/`; the 3-hourly report at `https://carlostoledo.co/reports/return-levels.html`. The hindcast is Accensi (2021); the buoy data are NDBC's and the wind–wave data the World Data Center for Climate's, both as distributed by the benchmark's repository; the design table is Bhaskaran et al. (2023) (CC BY 4.0).

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the author used large language models (Claude, Anthropic) to draft code, exposition and this manuscript in a certified-verification setting: constructions and prose were proposed with the help of the models, and every quantitative claim is decided by outward-rounded interval arithmetic or exact rational arithmetic, with adversarial controls that must fire for a record to be accepted. After using the tool the author reviewed and edited the content and takes full responsibility for the content of the publication. The records, not the models, are the evidence offered; they are public and re-runnable.

Funding

This work received no external funding.

Appendix A. The ten claims as read

#	quotation from Reis et al. (2026)	rule over the certified cells
1	"as the block size increased from daily to weekly and monthly, there was a marked reduction in the number of points classified as Exponentiated Weibull – the distribution that previously dominated – and a corresponding increase in the standard Weibull distribution"	the exponentiated Weibull decided at fewer cells, and the Weibull at more, at each step daily → weekly → monthly (Anderson–Darling)
2	"In the daily block, … the Generalized Gamma gains prominence along the coastal region south of South America"	the generalized gamma the most frequent decided choice in the box, daily maxima (Anderson–Darling)
3	"In the weekly block, the distinction between the northern, central, and southern ACC sectors weakens substantially, with the Generalized Gamma expanding its spatial influence"	the generalized gamma decided at more cells in the weekly block than in the daily (Anderson–Darling)
4	"By the monthly block, this diversity nearly disappears: the Exponentiated Weibull dominates most of the Southern Ocean, with only small residual areas of Gumbel"	the exponentiated Weibull the decided choice at more than half the certified cells, monthly maxima (Anderson–Darling)
5	"as the block size increases, a transition toward the standard Weibull distribution occurs"	the Weibull decided at more cells in the monthly block than in the daily (Anderson–Darling)
6	"in the monthly block, the Exponentiated Weibull again becomes dominant"	the exponentiated Weibull the decided choice at more than half the certified cells, monthly maxima (Anderson–Darling)
7	"Across most mid- and high-latitude ocean basins, including the North Atlantic, North Pacific, and Arabian Sea, a systematic reduction in estimated Hs values is observed as the block size increases"	at more than half the certified cells, the 100-year level of the monthly block's decided family lies wholly below the daily block's
8	"Across most mid- and high-latitude ocean basins, including the North Atlantic, North Pacific, and Arabian Sea, a systematic reduction in estimated Hs values is observed as the block size increases"	at more than half the certified cells, the 100-year level of the monthly block's decided family lies wholly below the daily block's
9	"The Arabian Sea displays a marked reduction in projected extremes as block size increases"	at more than half the certified cells, the 100-year level of the monthly block's decided family lies wholly below the daily block's
10	"In the western Pacific south of Japan, return levels increase for longer block sizes, particularly for the 1000-year period"	at more than half the certified cells, the 1000-year level of the monthly block's decided family lies wholly above the daily block's

Appendix B. Aggregation on every 3-hourly series

Table B.7: Reading only the family the Anderson–Darling statistic selects at each of the four paper blocks (and the annual block on the buoys), the lowest and highest 100-year level and their ratio.

series	lowest (block, family)	highest (block, family)	factor
buoy A	13.05 (monthly, lognormal)	17.38 (weekly, exp. Weibull)	1.33
buoy B	8.10 (weekly, Gumbel)	12.21 (native, lognormal)	1.51
buoy C	9.19 (weekly, exp. Weibull)	12.21 (native, gen. gamma)	1.33
campos	5.84 (monthly, gen. gamma)	6.13 (weekly, exp. Weibull)	1.05
santos	7.04 (daily, exp. Weibull)	7.62 (monthly, exp. Weibull)	1.08
espirito-santo	5.04 (monthly, gen. gamma)	5.77 (native, exp. Weibull)	1.15
pelotas	11.23 (daily, exp. Weibull)	11.51 (weekly, exp. Weibull)	1.02
potiguar	3.17 (monthly, gen. gamma)	3.32 (daily, gen. gamma)	1.05
foz-amazonas	4.61 (monthly, gen. gamma)	5.96 (native, lognormal)	1.29
drake	15.46 (weekly, gen. gamma)	17.45 (native, exp. Weibull)	1.13
acc-central	18.28 (daily, gen. gamma)	20.86 (monthly, Gumbel)	1.14
southern-high	17.71 (weekly, lognormal)	18.06 (monthly, exp. Weibull)	1.02
subantarctic-north	15.51 (weekly, lognormal)	18.60 (native, exp. Weibull)	1.20
arabian-sea	23.65 (monthly, exp. Weibull)	360.85 (native, exp. Weibull)	15.26
south-of-japan	17.84 (weekly, exp. Weibull)	28.88 (monthly, exp. Weibull)	1.62
north-atlantic	25.89 (weekly, gen. gamma)	28.48 (native, lognormal)	1.10

Appendix C. The benchmark's printed marginals, decided

Table C.8: The H_s marginals printed in the appendix of Haselsteiner et al. (2021) for the provided ten years of buoys A, B and C, decided on those hours, with the hourly 100-year level over the box the printed digits allow.

contribution	buoy	family	printed	decided	100-yr H_s (m
1 & 2	A	Weibull	0.944, 1.48, 0.0981	location at the smallest hour to the digits; undecided	5.59–5.66
1 & 2	B	Weibull	1.14, 1.60, 0.188	location at the smallest hour to the digits; undecided	5.98–6.09
1 & 2	C	Weibull	1.16, 1.56, 0.0566	location at the smallest hour to the digits; undecided	6.20–6.32
3	A	3-p lognormal	0.717, 0.635, 0.0634	a certified local maximum of an unbounded likelihood	14.44–14.53
3	B	3-p lognormal	0.972, 0.572, 0.0633	a certified local maximum of an unbounded likelihood	14.54–14.62
3	C	3-p lognormal	0.937, 0.620, -0.0318	a certified local maximum of an unbounded likelihood	17.48–17.58
4	A	exp. Weibull	0.207, 0.684, 7.79	below the certified maximum by 877 or more (least squares by design)	11.57–11.70
4	B	exp. Weibull	0.0988, 0.584, 36.6	not maximum likelihood (weighted least squares)	12.94–13.06
4	C	exp. Weibull	0.227, 0.697, 9.85	not maximum likelihood (weighted least squares)	12.03–12.15
8	A	Weibull	1.0651, 1.6399	the rounding of the certified MLE	5.25–5.25
8	B	Weibull	1.3654, 1.8893	the rounding of the certified MLE	5.45–5.45
8	C	Weibull	5.0475, 5.5633	the certified Weibull MLE of the zero-up-crossing period, not of H_s	8.08–8.08
9	A	Weibull	0.4983, 0.8573, 0.4187	10,426 hours below the printed location	10.96–10.96
9	B	Weibull	0.6539, 0.9710, 0.5658	10,904 hours below the printed location	10.24–10.24
9	C	Weibull	0.7291, 1.0134, 0.3910	9,410 hours below the printed location	10.03–10.03

References

Abramowitz, M., Stegun, I.A., 1964. Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables. National Bureau of Standards, Applied Mathematics Series 55, Washington.

Accensi, M., 2021. GLOBMULTI_ERA5_GLOBCUR_01. IFREMER. https://doi.org/10.12770/857a3337-f59a-481a-bf98-5561e8b61e7b

Alday, M., Accensi, M., Ardhuin, F., Dodet, G., 2021. A global wave parameter database for geophysical applications. Part 3: Improved forcing and spectral resolution. Ocean Modelling 166, 101848.

Anderson, T.W., Darling, D.A., 1954. A test of goodness of fit. Journal of the American Statistical Association 49, 765–769.

Bhaskaran, S.K., Verma, A.S., Goupee, A.J., Bhattacharya, S., Nejad, A.R., Shi, W., 2023. Comparison of extreme wind and waves using different statistical methods in 40 offshore wind energy lease areas worldwide. Energies 16, 6935.

Coles, S., 2001. An Introduction to Statistical Modeling of Extreme Values. Springer, London.

DNV, 2021. DNV-RP-C205: Environmental conditions and environmental loads. Recommended practice.

Fortune, S., Van Wyk, C.J., 1996. Static analysis yields efficient exact integer arithmetic for computational geometry. ACM Transactions on Graphics 15, 223–248.

Hansen, E., Walster, G.W., 2004. Global Optimization Using Interval Analysis, 2nd ed. Marcel Dekker, New York.

Harper, W.V., Eschenbach, T.G., James, T.R., 2011. Concerns about maximum likelihood estimation for the three-parameter Weibull distribution: case study of statistical software. The American Statistician 65, 44–54.

Haselsteiner, A.F., Ohlendorf, J.-H., Wosniok, W., Thoben, K.-D., 2017. Deriving environmental contours from highest density regions. Coastal Engineering 123, 42–51.

Haselsteiner, A.F., Coe, R.G., Manuel, L., Chai, W., Leira, B., Clarindo, G., Guedes Soares, C., Hannesdóttir, Á., Dimitrov, N., Sander, A., Ohlendorf, J.-H., Thoben, K.-D., de Hauteclocque, G., Mackay, E., Jonathan, P., Qiao, C., Myers, A., Rode, A., Hildebrandt, A., Schmidt, B., Vanem, E., Huseby, A.B., 2021. A benchmarking exercise for environmental contours. Ocean Engineering 236, 109504.

Higham, N.J., 2002. Accuracy and Stability of Numerical Algorithms, 2nd ed. SIAM, Philadelphia.

Hill, B.M., 1963. The three-parameter lognormal distribution and Bayesian analysis of a point-source epidemic. Journal of the American Statistical Association 58, 72–84.

Hormann, K., Agathos, A., 2001. The point in polygon problem for arbitrary polygons. Computational Geometry 20, 131–144.

Hunter, J.D., 2007. Matplotlib: A 2D graphics environment. Computing in Science & Engineering 9, 90–95.

ISO, 2015. ISO 19901-1:2015 Petroleum and natural gas industries — Specific requirements for offshore structures — Part 1: Metocean design and operating considerations.

Jenkinson, A.F., 1955. The frequency distribution of the annual maximum (or minimum) values of meteorological elements. Quarterly Journal of the Royal Meteorological Society 81, 158–171.

Kettner, L., Mehlhorn, K., Pion, S., Schirra, S., Yap, C., 2008. Classroom examples of robustness problems in geometric computations. Computational Geometry 40, 61–78.

Krawczyk, R., 1969. Newton-Algorithmen zur Bestimmung von Nullstellen mit Fehlerschranken. Computing 4, 187–201.

Mackay, E., Haselsteiner, A.F., 2021. Marginal and total exceedance probabilities of environmental contours. Marine Structures 75, 102863.

Moore, R.E., Kearfott, R.B., Cloud, M.J., 2009. Introduction to Interval Analysis. SIAM, Philadelphia.

Mudholkar, G.S., Srivastava, D.K., 1993. Exponentiated Weibull family for analyzing bathtub failure-rate data. IEEE Transactions on Reliability 42, 299–302.

Ogita, T., Rump, S.M., Oishi, S., 2005. Accurate sum and dot product. SIAM Journal on Scientific Computing 26, 1955–1988.

Prentice, R.L., 1974. A log gamma model and its maximum likelihood estimation. Biometrika 61, 539–544.

Reis, R.A.N., Guimarães, P.V., Farina, L., Paul, S., de Paula, T.P., Ribeiro, E.O., 2026. Global assessment of extreme value analysis for significant wave height. Ocean Engineering 359, 125841. Corrigendum: Ocean Engineering 361, 126267.

Rump, S.M., 2010. Verification methods: rigorous results using floating-point arithmetic. Acta Numerica 19, 287–449.

Shewchuk, J.R., 1997. Adaptive precision floating-point arithmetic and fast robust geometric predicates. Discrete & Computational Geometry 18, 305–363.

Shimrat, M., 1962. Algorithm 112: Position of point relative to polygon. Communications of the ACM 5, 434.

Smith, R.L., 1985. Maximum likelihood estimation in a class of nonregular cases. Biometrika 72, 67–90.

Smith, R.L., Weissman, I., 1994. Estimating the extremal index. Journal of the Royal Statistical Society, Series B 56, 515–528.

Stacy, E.W., 1962. A generalization of the gamma distribution. The Annals of Mathematical Statistics 33, 1187–1192.

Toledo, C., 2026a. The cert-machine method: three-valued decisions with witnesses in exact and interval arithmetic, and the verification discipline that keeps them sound. Draft for review, version 0.1 of 2026-10-09. `https://carlostoledo.co/paper/cert-machine-method.pdf`

Toledo, C., 2026b. The environmental-contour benchmark re-decided exactly: 150 submitted contours as exact polygons, 1.8 million hourly sea states classified in exact arithmetic, and the counts the organizers printed. Draft for review, version 0.1 of 2026-10-01. `https://carlostoledo.co/paper/ec-benchmark.pdf`

Tucker, W., 2011. Validated Numerics: A Short Introduction to Rigorous Computations. Princeton University Press, Princeton.

Virtanen, P., Gommers, R., Oliphant, T.E., et al., 2020. SciPy 1.0: fundamental algorithms for scientific computing in Python. Nature Methods 17, 261–272.

Winterstein, S.R., Ude, T.C., Cornell, C.A., Bjerager, P., Haver, S., 1993. Environmental parameters for extreme response: inverse FORM with omission factors. In: Proceedings of the 6th International Conference on Structural Safety and Reliability (ICOSSAR 93), Innsbruck.

Yap, C.K., 1997. Towards exact geometric computation. Computational Geometry 7, 3–23.